

Exp No: 24 Modeling Clock-Driven Concurrent Processes Using SC_THREAD and wait()
in SystemC
DATE: 25/08/2026

```
kaviya@KV: ~
GNU nano 8.7.1
#include <systemc.h>

SC_MODULE(ClockExample) {
    sc_in<bool> clk;

    void process() {
        while (true) {
            wait();
            std::cout << "Time: " << sc_time_stamp()
            << " - Clock Triggered" << std::endl;
        }
    }

    SC_CTOR(ClockExample) {
        SC_THREAD(process);
        sensitive << clk.pos();
    }
};

int sc_main(int argc, char* argv[]) {
    sc_clock clock("clock", 1, SC_NS);

    ClockExample module("module");
    module.clk(clock);

    sc_start(5, SC_NS);

    return 0;
}
```

kaviya@KV: ~

X + v

```
aviya@KV:~$ nano exp24.cpp
```

```
aviya@KV:~$ g++ -o test_systemc test_systemc.cpp -lsystemc
```

```
c1plus: fatal error: test_systemc.cpp: No such file or directory  
compilation terminated.
```

```
aviya@KV:~$ g++ -o exp24 test_systemc.cpp -lsystemc
```

```
c1plus: fatal error: test_systemc.cpp: No such file or directory  
compilation terminated.
```

```
aviya@KV:~$ g++ -o exp24 exp24.cpp -lsystemc
```

```
aviya@KV:~$ ./exp24
```

```
SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
```

```
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```
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```

```
ime: 0 s - Clock Triggered
```

```
ime: 1 ns - Clock Triggered
```

```
ime: 2 ns - Clock Triggered
```

```
ime: 3 ns - Clock Triggered
```

```
ime: 4 ns - Clock Triggered
```

```
aviya@KV:~$ |
```